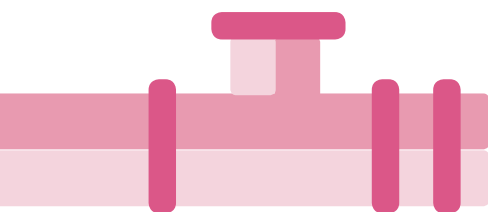


Using Apache Beam to Pipeline and Process Data

We live in an era where data is growing at a huge pace and will continue to do so as new technologies evolve. This data needs to be pipelined in order to be managed and analysed. Apache Beam helps with that.



Across the globe, tons of data is being generated every moment, thanks to:

- Social communication networks
 - E-commerce platforms
 - Supply chain management
 - Stock markets
 - Maps/location data
 - Fraud detection in banking and finance
 - Log files monitoring and analysis
 - Electronic health records
 - E-governance data sets
- To process such data effectively,

there is a need to integrate data pipelines so that congestion-free communication can take place.

Data flow pipelines and batch processing

The term data pipeline refers to the effective and rapid process of transporting data from its source to its destination (including the data warehouse). Data undergoes several transformations and optimisations along the road, ultimately arriving in a form suitable for analysis and business insights.

Data pipelining has many stages for real-time applications. Data is first ingested at the start of the pipeline. After that, there is a sequence of operations, each of which produces an output that serves as an input for the subsequent operation. This procedure is repeated until the pipeline is finished. It's possible to execute separate processes simultaneously.

The transmission of data from one system to another is fraught with the potential for errors or slowdowns due to the complexity of the networks involved. The extent and effect of the issues only grow as the breadth and depth of the data's function expands. For this reason, data pipelines are crucial. They allow seamless, automatic transfer of data from one phase to the next, reducing the need for any human intervention. They are crucial for real-time analytics, which allow you to act on information more quickly.

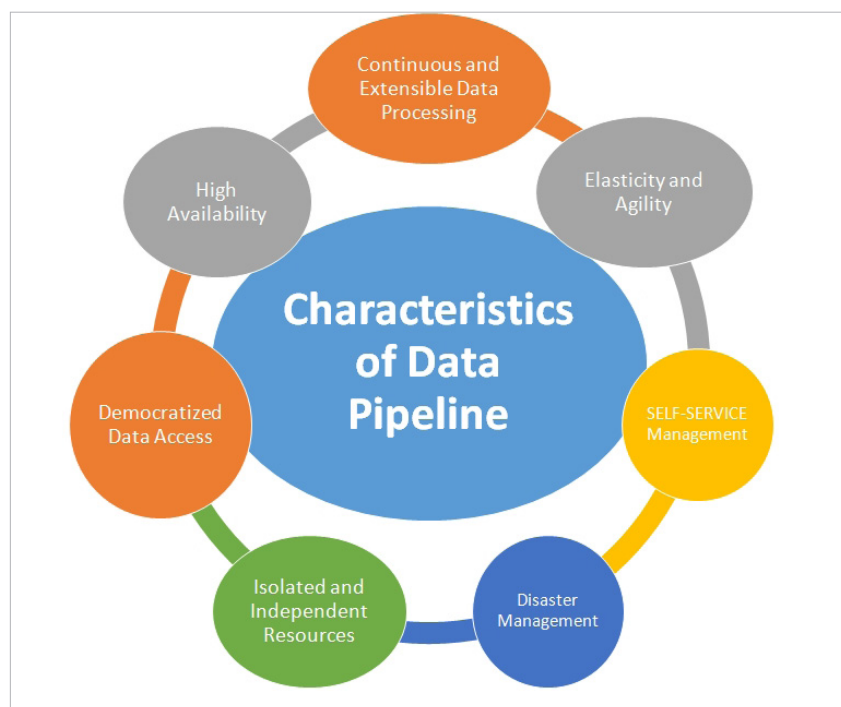


Figure 1: Key characteristics of a data pipeline